University
Mohammed VI
Polytechnic

College of
Computing

Deliverable # 6: Lab6 (e.g.: Conceptual Design ....)

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering

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Team Information

Team Name	UNO
Member 1	Daniel Tchonang
Member 2	Amine Sira
Member 3	Mohammed Oumane
Member 4	Zaid Zehari
Repository Link	`https://github.com/unocomputing/unodatabase.git`

1 Roles

- Daniel: SQL + report
- Sira: SQL + report
- Mohammed: SQL
- Zaid; SQL

2 Difficulties Encountered

None.

3 Justification of the ON DELETE choices

In most cases we would have wanted `SET NULL`, but the foreign key column is either `NOT NULL` or part of the primary key, so it cannot be set to `NULL`. In these cases we used `NO ACTION`: the deletion is rejected as long as other rows still reference the deleted row.

CASCADE

- `FundsPrj` and `FundsLab`: these tables only record a link. If the budget, the project or the laboratory is deleted, the link has no meaning, so it is deleted too (the other entity is not).
- `modelId` in `EquipmentUnit`: a unit belongs to exactly one equipment model (EQ5), so it cannot exist without it.
- `Requires`: if the certification or the equipment model does not exist, the requirement does not exist either.
- `Holds`: a person who does not exist cannot hold anything, and a certification that does not exist cannot be held.
- `serialNo` in `Maintenance` and `CalibrationRecord`: both are weak entities of `EquipmentUnit` (MA2, CA4), so they cannot exist if the equipment unit is deleted.
- `madeByPid` and `projectCode` in `Reservation`: a reservation must be made by one person for one project (RE1), so it cannot remain if either is deleted.
- `Reserves`: if the reservation is deleted, the log about its equipment is deleted, and likewise if the equipment unit is deleted.

NO ACTION

- `Attached` and `Participates`: both columns are part of the primary key, so `SET NULL` is impossible. We also do not want to silently lose who worked in which laboratory or project, and with which role.

- `technicalPid` in `Maintenance`: each maintenance is carried out by exactly one technical staff member (MA3), so the column is `NOT NULL`. Deleting a technician must not erase the maintenance history.
- `labId` in `EquipmentUnit`: a unit always has a current location (EQ3), so the column is `NOT NULL`. Even if the laboratory is deleted, the unit has still been bought, so it must be moved to another laboratory first.
- `Supplies`: all its foreign keys are part of the primary key, and we do not want to silently lose the negotiated prices.
- `Stocks`: `labId` and `consId` are part of the primary key, and `technicalPid` is `NOT NULL` because each stock is monitored by exactly one technician (CO8). A technician must be replaced before being deleted.
- `Consumes`: it records what each reservation consumed from which stock (CO9–CO10), so deleting a reservation or a stock must not silently falsify this history.

SET NULL

- `approverPid` in `Reservation`: the column is nullable because only some reservations require approval (RE3). If the approver is deleted, the reservation and its equipment must remain, so only the link to the approver is removed.

4 Question 4

4.1 Every research project must have exactly one PI (PR3)

- Code concerned: ResearchProject and Participates tables.
- The primary key prevents the same person from participating in the same project more than once, but it cannot guarantee that every project has exactly one PI. The database must check the number of participants with the role PI whenever a participation is inserted, updated, or deleted.

4.2 Every laboratory must be funded by at least one budget (BD3)

- Code concerned: Laboratory and FundsLab tables.
- The foreign keys guarantee that each funding record refers to an existing budget and laboratory, but they cannot guarantee that every laboratory has at least one funding record. The database must check whether a laboratory has associated funding when a laboratory is created or its last funding record is deleted.

4.3 Administrative staff cannot participate in research projects (PR2)

- Code concerned :Administrative and Participates tables.

- The foreign key checks that the participant exists in Person, but it does not check whether that person belongs to the Administrative table. The database must verify the person's employee subtype whenever a participation record is inserted or updated.

4.4 Every reservation must reserve at least one equipment unit (RE4)

- Code concerned :Reservation and Reserves tables.
- The foreign keys guarantee that each reservation-to-equipment association refers to existing records, but they cannot require every reservation to have at least one associated equipment unit. The database must check whether a reservation has an associated record in Reserves when a reservation is created or its last equipment association is deleted.